100-Species Animal Classifier: Fine-Tuned CNNs and CLIP Embeddings for Wildlife Identification

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https://github.com/ViratGarg2/smai_project

Abstract

We present a comparative study of three strategies for 100-class animal species classification across mammals, birds, and butterflies. **Model 1** fine-tunes a single ResNet50 directly on all 100 classes, reaching **97.22%** test accuracy. **Model 2** employs a two-stage hierarchical pipeline — a group classifier routes each image to a dedicated species model — achieving 96.89% end-to-end accuracy. **Model 3** requires no gradient-based training on the target dataset; instead, frozen CLIP/BioCLIP embeddings are paired with KNN and SVM classifiers, reaching up to 96.61% with BioCLIP + SVM. PCA analysis of the embedding space reveals strong macro-level separation across groups (silhouette = 0.14) but significant intra-group overlap for butterflies (0.06), explaining their comparatively lower performance. A live Streamlit demo is provided for interactive top-3 species prediction.

1. Introduction

Automated wildlife identification has practical value in conservation monitoring, ecotourism, and biodiversity research. This work builds a species-level classifier capable of distinguishing 100 animal species from a single image, spanning mammals (45 species), birds (25 species), and butterflies (30 species). We evaluate three progressively different classification paradigms: direct end-to-end fine-tuning, hierarchical decomposition, and training-free embedding-based classification.

Our contributions are: (i) a unified 100-class benchmark combining three public datasets, (ii) a systematic comparison of fine-tuned CNN pipelines against classical classifiers applied to modern visual embeddings, and (iii) an analysis of CLIP embedding geometry using PCA and silhouette scores to explain per-group performance differences.

2. Dataset

The dataset is a custom compilation merging three publicly available animal image datasets into a unified 100-class label space. It is hosted on HuggingFace at ViratGarg/animal_species_SMAI.

Table 1: Dataset composition.

Subset	Species	Images
Mammals	45	13,751
Birds	25	8,750
Butterflies	30	4,158
Total	**100**	**26,659**

The data is split 80/10/10 into train (21,327), validation (2,666), and test (2,666) images. Per-species test support ranges from approximately 11 to 36 samples. Standard ImageNet normalization is applied; all images are resized to 224×224.

3. Methods

3.1. Model 1: Direct ResNet50 Fine-Tuning

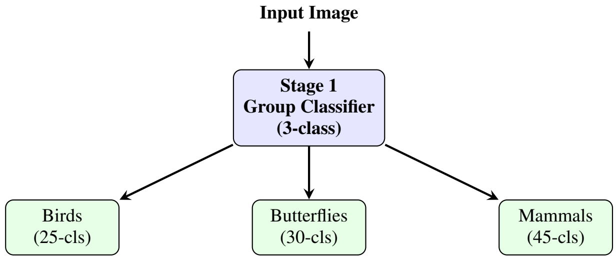
A ResNet50 pretrained on ImageNet1K [1] serves as the backbone. The final fully-connected layer is replaced with a two-component head: Dropout($p = 0.4$) followed by a linear projection to 100 output classes. The full network is trained end-to-end with the hyperparameters in Table 2. This flat approach learns any cross-category visual features without imposing structural priors.

Table 2: Model 1 hyperparameters.

Parameter	Value
Optimizer	AdamW
Learning rate	1×10^{-4}
Weight decay	1×10^{-4}
Epochs	20
LR schedule	Cosine Annealing ($T_{\max} = 20$)
Loss	Cross-Entropy ($\varepsilon = 0.1$ label smoothing)
Dropout	0.4
Batch size	32

3.2. Model 2: Two-Stage Hierarchical Pipeline

Four ResNet50 models are arranged in a cascaded pipeline. *Stage 1* is a 3-way group classifier (birds / butterflies / mammals). At test time, the predicted group label routes the image to one of three *Stage 2* species classifiers, reducing the per-model classification problem from 100-way to at most 45-way. All four models share the same hyperparameters (same as Table 2 but 15 epochs), with checkpointing on best validation accuracy.



3.3. Model 3: CLIP Embeddings + Classical Classifiers

No gradient-based training on the target dataset is required. Three frozen encoders [2, 3] are evaluated:

- **CLIP ViT-B/16** — general-purpose, 400M image-text pairs.
- **CLIP ViT-B/32** — lighter general-purpose variant.
- **BioCLIP** — domain-specialized for biological taxonomy.

Three classification heads are applied to the extracted embeddings: *Zero-Shot* (cosine similarity to text prompts of the form *"a photo of a [species name]"*), *KNN* ($k = 1$–20), and *Linear SVM*. The choice of KNN and SVM is deliberate — these are classical algorithms covered in the course curriculum, and pairing them with CLIP embeddings demonstrates that strong representations can unlock competitive performance even with simple classifiers.

4. Feature Space Analysis

PCA projections of CLIP ViT-B/16 embeddings onto the first two principal components (Figure 1) reveal clear macro-level separation: butterflies form a compact, well-separated cluster while mammals and birds overlap at the boundary. Silhouette scores (Table 3) quantify within-group species separation; butterflies score the lowest (0.06), foreshadowing their harder classification.

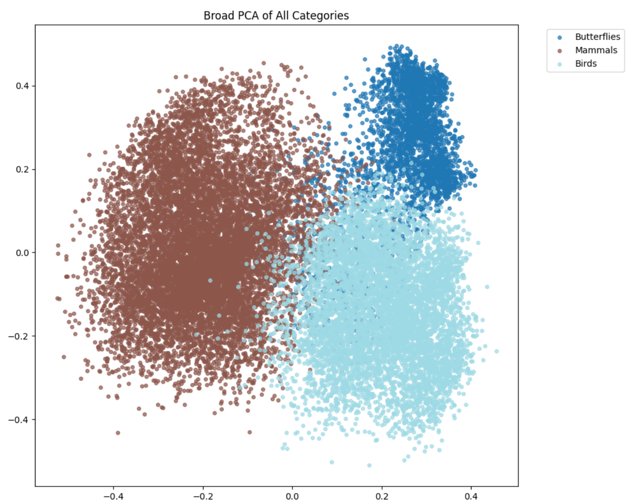


Figure 1: PCA of CLIP ViT-B/16 embeddings — all three animal groups. Butterflies (green) cluster distinctly; mammals and birds overlap moderately.

Table 3: CLIP embedding silhouette scores.

Scope	Silhouette Score
3 broad categories	0.1449
Mammals (45 species)	0.1165
Birds (25 species)	0.1192
Butterflies (30 sp.)	0.0600

5. Results

5.1. Model 1: Single ResNet50

Table 4: Model 1 training curve (selected epochs).

Epoch	Train	Val	Test
1	78.09%	94.37%	94.11%
5	98.18%	96.06%	95.87%
10	99.34%	96.70%	96.51%
15	99.84%	97.00%	97.11%
20	99.96%	97.41%	**97.22%**

The model converges rapidly — validation accuracy exceeds 94% after a single epoch. A stable generalization gap of ≈2.5% persists through training, controlled by label smoothing and cosine annealing. Final test accuracy: **97.22%**; macro specificity: **99.97%**.

5.2. Model 2: Two-Stage Hierarchical Pipeline

The Stage 1 group classifier achieves **99.89%** accuracy (macro F1: 99.86%), with only 3 out of 2,666 test images misrouted. Misrouted samples cannot be recovered by Stage 2 regardless of its performance. Table 5 reports Stage 2 species accuracies and the end-to-end pipeline performance.

Table 5: Model 2 — Stage 2 and end-to-end results.

Group / Metric	Acc.	Macro F1
Birds (25 classes)	97.14%	97.13%
Butterflies (30 classes)	97.59%	97.54%
Mammals (45 classes)	96.66%	96.59%
End-to-end species acc.	**96.89%**	96.91%
End-to-end macro spec.	99.97%	—

5.3. Model 3: CLIP Embeddings

SVM results. Table 6 shows that BioCLIP substantially outperforms CLIP ViT-B/16 for butterflies (91.71% → 98.08%), consistent with its biological taxonomy pretraining.

Table 6: SVM classification on CLIP embeddings.

Model	Mam.	Birds	Butt.	All
CLIP ViT-B/16	95.86	97.54	91.71	95.76
BioCLIP	94.62	**99.03**	**98.08**	**96.61**

Zero-shot results. Table 7 reports zero-shot performance using cosine similarity to text prompts. Butterfly accuracy is severely degraded for standard CLIP models (37–39%), while BioCLIP recovers to 55.17%, confirming that the bottleneck is text-prompt matching quality rather than the image encoder.

Table 7: Zero-shot classification accuracy (%).

Model	Mam.	Birds	Butt.	All
CLIP ViT-B/32	92.26	79.09	37.86	78.72
CLIP ViT-B/16	93.82	83.49	39.42	81.14
BioCLIP	81.46	**89.71**	**55.17**	80.01

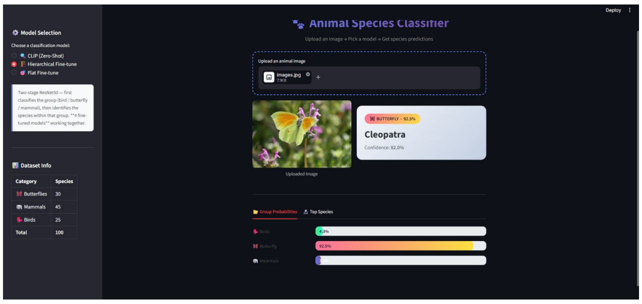
5.4. Summary Comparison

Table 8: Full dataset test accuracy — all methods.

Method	Test Acc.
Model 1: Single ResNet50	**97.22%**
Model 2: Hierarchical ResNet50	96.89%
Model 3: BioCLIP + SVM	96.61%
Model 3: CLIP ViT-B/16 + SVM	95.76%
Model 3: CLIP + KNN (best k)	≈94–95%
Zero-Shot CLIP ViT-B/16	81.14%
Zero-Shot BioCLIP	80.01%

6. Demo Application

We built a Streamlit web application for interactive species prediction. Users upload a single image; the app runs inference through all three model families and returns top-3 predicted species with confidence scores. For Model 2, the routing decision (group classification) is also displayed. Figure 2 shows screenshots of the deployed application.



(a) Top-3 species predictions with confidence scores.

Figure 2: Streamlit demo — showcase larger images within the single column (now touching).

7. Discussion

Why are accuracies uniformly high? ResNet50 pretrained on ImageNet1K has processed 1.2M images across 1,000 diverse categories, and CLIP was trained on 400M image-text pairs — both are substantially overqualified for 100-class animal classification. The models already encode feature detectors for fur texture, wing patterns, and body shape; fine-tuning merely aligns the final head to the new label space. Moreover, most species pairs in this dataset are visually highly distinctive (a giraffe vs. a warthog), unlike fine-grained benchmarks such as CUB-200 where inter-class differences are subtle.

Why does CLIP underperform fine-tuned ResNets? KNN and SVM operate on a 512-dimensional embedding space not explicitly optimized for species-level separability. Fine-tuned ResNets reshape their entire representation hierarchy to maximize within-dataset discrimination. That said, the use of KNN and SVM was deliberately chosen to integrate classical course-taught algorithms with modern vision embeddings, demonstrating that strong representations remain competitive even with simple classifiers.

Butterflies are the hardest group. Low silhouette score (0.06), weakest zero-shot accuracy (≈39%), and lowest KNN accuracy (≈88%) consistently identify butterflies as the most challenging subset. BioCLIP's large improvement (91.71% → 98.08% on SVM) demonstrates this is a representation quality issue — domain-specific pretraining data matters significantly for entomological classification.

8. Limitations

Among mammals, *seal* (F1: 0.828) and *blue_whale* (F1: 0.840) are the hardest individual classes. Seals share visual features with sea lions present in the dataset, and blue whales lack distinctive close-up textures compared to terrestrial species. With only ≈26 test samples per species on average, per-class F1 estimates carry high variance. No ablation over augmentation strategies (mixup, stronger color jitter) was conducted, which is a direction for future work.

9. Conclusion

All three classification approaches exceed 95% accuracy on the 100-class combined benchmark. Direct ResNet50 fine-tuning achieves the best overall accuracy (97.22%), the hierarchical pipeline (96.89%) offers superior modularity and interpretability, and BioCLIP + SVM (96.61%) demonstrates that training-free embedding classifiers remain competitive when domain-appropriate pretrained models are used. High accuracy across methods is attributable to the strength of modern pretrained representations applied to a visually distinctive, well-curated dataset.

Acknowledgements

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